

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285128

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Chauhan; Chet

INTEGRATING AND MANAGING SOCIAL NETWORKING INFORMATION IN AN ON-DEMAND DATABASE SYSTEM

Abstract

Some embodiments comprise integrating information from a social network into a multi-tenant database system. A plurality of information from the social network is retrieved, using a processor and a network interface of a server computer in the multi-tenant database system, wherein the plurality of information is associated with a message transmitted using the social network. Metadata related to the transmitted message is generated, using the processor. A conversation object is generated, using the processor, based on the plurality of information associated with the transmitted message and the metadata related to the transmitted message. The conversation object is then stored in an entity in the multi-tenant database system, using the processor of the server computer.

Inventors: Chauhan; Chet (San Francisco, CA)

Applicant: salesforce.com, inc. (San Francisco, CA)

Family ID: 1000008619524

Appl. No.: 19/216102

Filed: May 22, 2025

Related U.S. Application Data

parent US continuation 18746365 20240618 parent-grant-document US 12354113 child US 19216102

parent US continuation 18331563 20230608 parent-grant-document US 12045834 child US 18746365

parent US continuation 17806997 20220615 parent-grant-document US 11715110 child US 18331563

parent US continuation 17249956 20210319 parent-grant-document US 11386434 child US 17806997

parent US continuation 16676182 20191106 parent-grant-document US 10977661 child US

17249956
parent US continuation 15374807 20161209 parent-grant-document US 10489790 child US
16676182
parent US continuation 14145870 20131231 parent-grant-document US 9519711 child US
15374807
parent US continuation 12917326 20101101 parent-grant-document US 8621015 child US
14145870
us-provisional-application US 61256861 20091030

Publication Classification

Int. Cl.: **G06Q30/0201** (20230101); **G06F16/95** (20190101); **G06F16/9535** (20190101);
G06F16/958 (20190101); **G06Q50/00** (20240101); **H04L51/216** (20220101);
H04L51/52 (20220101); **H04L67/10** (20220101)

U.S. Cl.:

CPC **G06Q30/0201** (20130101); **G06F16/95** (20190101); **G06F16/9535** (20190101);
G06F16/958 (20190101); **G06Q50/01** (20130101); **H04L51/216** (20220501);
H04L51/52 (20220501); **H04L67/10** (20130101);

Background/Summary

INCORPORATION BY REFERENCE

[0001] An Application Data Sheet is filed concurrently with this specification as part of the present application. Each application that the present application claims benefit of or priority to as identified in the concurrently filed Application Data Sheet is incorporated by reference herein in its entirety and for all purposes.

BACKGROUND

[0002] Social networks, such as Twitter®, Facebook®, and Yammer® have grown in popularity not only for casual social purposes, but also for work-related purposes. Users of social networks post messages related to a company's products and/or services containing information that could be useful to understand customers, track market developments, manage public relations, and address issues related to the company's products and/or services, employees, or the company itself.

SUMMARY

[0003] The implementations disclosed herein relate generally to database systems and more particularly to integration of information from social networks into multi-tenant database systems ("MTS"). Embodiments of the implementations are particularly useful for integrating information from social networks such as twitter.com, yammer.com, and other similar services. The implementations described herein provide methods and systems for integrating information harvested from social networks (e.g., posted messages) into a MTS. Various embodiments provide methods and systems for enabling customer support representatives to manage social networks like any other support channel. Various embodiments provide methods and systems for enabling multiple agents to pull, filter, process, search through, and take action based on such harvested information. Various embodiments provide methods and systems for using templates for such harvested information and storing harvested information in a knowledge base to increase productivity.

[0004] Some embodiments comprise integrating information from a social network into a multi-

tenant database system. A plurality of information from the social network is retrieved, using a processor and a network interface of a server computer in the multi-tenant database system, wherein the plurality of information is associated with a message transmitted using the social network. Metadata related to the transmitted message is generated, using the processor. A conversation object is generated, using the processor, based on the plurality of information associated with the transmitted message and the metadata related to the transmitted message. The conversation object is then stored in an entity in the multi-tenant database system, using the processor of the server computer.

[0005] Some embodiments comprise a non-transitory computer-readable medium tangibly encoded with instructions for performing the above-described operations and variations thereof.

[0006] Some embodiments comprise a system comprising a processor, a network interface, and a non-transitory computer-readable medium tangibly encoded with instructions for performing the above-described operations and variations thereof.

[0007] Reference to the remaining portions of the specification, including the drawings and claims, will realize other features and advantages of the implementations. Further features and advantages of the implementations, as well as the structure and operation of various embodiments, are described in detail below with respect to the accompanying drawings. In the drawings, like reference numbers indicate identical or functionally similar elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic illustrating an environment including a MTS.

[0009] FIG. 2 is a schematic illustrating elements of a MTS and interconnections therein in an embodiment.

[0010] FIG. 3 is a schematic illustrating an example of objects represented as a main table in an embodiment.

[0011] FIG. 4 is a schematic illustrating an example system for integrating and managing social networking information in a MTS in an embodiment.

[0012] FIG. 5 illustrates an example process flow for an integration with the twitter social network.

DETAILED DESCRIPTION

[0013] The implementations described herein provide methods and systems for integrating information harvested from social networks (e.g., posted messages) into a MTS. Customer support representatives are able to manage social networks like any other support channel. Multiple agents can pull, filter, process, search through, and take action based on such harvested information. Templates can be used to characterize harvested information and store it into standard object types. Harvested information can be stored in a knowledge base to increase productivity.

Definitions

[0014] Each database can generally be viewed as a collection of objects, such as a set of logical tables, containing data fitted into predefined categories. A table, one representation of a data object, is used herein to simplify the conceptual description of objects and custom objects in the present disclosure. It should be understood that the terms table, object and entity may be used interchangeably herein. Each table generally contains one or more data categories logically arranged as columns or fields in a viewable schema. Each row or record of a table contains an instance of data for each category defined by the fields. For example, a CRM database may include a table that describes a customer with fields for basic contact information such as name, address, phone number, fax number, etc. Another table might describe a purchase order, including fields for information such as customer, product, sale price, date, etc.

[0015] As used herein, the term multi-tenant database system ("MTS") refers to those systems in

which various elements of hardware and software of the database system may be shared between one or more customers. For example, a given application server may simultaneously process requests for a great number of customers, and a given database table may store rows for a potentially much greater number of customers. In some MTS embodiments, standard entity tables might be provided. For CRM database applications, such standard entities might include tables for relevant data, such as Account, Contact, Lead and Opportunity, each containing predefined fields. [0016] As used herein, the term social network or social networking site refers to any multiuser online system that enables its users to create and preserve connections, communicate by posting broadcast or private messages, and share information with other users of the online system. Examples of social networks include twitter, Facebook, and Yammer.

[0017] As used herein, the term message includes any message posted on a social network, including but not limited to status updates, private messages sent to one or more users of the social network, or public messages for broadcast to everyone or to a subgroup of users.

Multi-Tenant Database System (MTS) Overview

[0018] FIG. 1 illustrates an environment wherein a MTS might be used. As illustrated in FIG. 1 (and in more detail in FIG. 2) any user systems 12 might interact via a network 14 with a MTS 16. The users of those user systems 12 might be users in differing capacities and the capacity of a particular user system 12 might be entirely determined by the current user. For example, when a salesperson is using a particular user system 12 to interact with MTS 16, that user system has the capacities allotted to that salesperson. However, while an administrator is using that user system to interact with MTS 16, that user system has the capacities allotted to that administrator. Network 14 can be a local area network ("LAN"), wide area network ("WAN"), wireless network, point-to-point network, star network, token ring network, hub network, or other configuration. As the most common type of network in current use is a Transfer Control Protocol and Internet Protocol ("TCP/IP") network such as the global internetwork of networks often referred to as the "Internet" with a capital "I," that will be used in many of the examples herein, but it should be understood that the networks that the system might use are not so limited, although TCP/IP is the currently preferred protocol.

[0019] User systems 12 might communicate with MTS 16 using TCP/IP and, at a higher network level, use other common Internet protocols to communicate, such as Hypertext Transfer Protocol ("HTTP"), file transfer protocol ("FTP"), Andrew File System ("AFS"), wireless application protocol ("WAP"), etc. As an example, where HTTP is used, user system 12 might include a HTTP client commonly referred to as a "browser" for sending and receiving HTTP messages from a HTTP server at MTS 16. Such a HTTP server might be implemented as the sole network interface between MTS 16 and network 14, but other techniques might be used as well or instead. In some embodiments, the interface between MTS 16 and network 14 includes load-balancing functionality, such as round-robin HTTP request distributors to balance loads and distribute incoming HTTP requests evenly over a plurality of HTTP servers. Preferably, each of the plurality of HTTP servers has access to the MTS's data, at least as for the users that are accessing that server.

[0020] In aspects, the system shown in FIG. 1 implements a web-based customer relationship management ("CRM") system. For example, in one aspect, MTS 16 can include application servers configured to implement and execute CRM software applications as well as provide related data, program code, forms, web pages and other information to and from user systems 12 and to store to, and retrieve from, a database system related data, objects and web page content. With a MTS, tenant data is preferably arranged so that data of one tenant is kept separate from that of other tenants so that one tenant does not have access to another's data, unless such data is expressly shared.

[0021] One arrangement for elements of MTS 16 is shown in FIG. 1, including a network interface 20, storage 22 for tenant data, storage 24 for system data accessible to MTS 16 and possibly multiple tenants, program code 26 for implementing various functions of MTS 16, and a process

space **28** for executing MTS system processes and tenant-specific processes, such as running applications as part of an application service.

[0022] Some elements in the system shown in FIG. **1** include conventional, well-known elements that need not be explained in detail here. For example, each user system **12** could include a desktop personal computer, workstation, laptop, personal digital assistant (“PDA”), cell phone, or any WAP-enabled device or any other computing device capable of interfacing directly or indirectly to the Internet or other network connection. User system **12** typically runs a HTTP client, e.g., a browsing program, such as Microsoft's Internet Explorer® browser, Mozilla's Firefox® browser, Netscape's Navigator® browser, Apple's Safari® browser, the Opera© browser, or a WAP-enabled browser in the case of a cell phone, PDA, or other wireless device, or the like, allowing a user (e.g., subscriber of a CRM system) of user system **12** to access, process and view information and pages available to it from MTS **16** over network **14**. Each user system **12** also typically includes one or more user interface devices, such as a keyboard, a mouse, touch screen, pen or the like, for interacting with a graphical user interface (“GUI”) provided by the browser on a display (e.g., monitor screen, LCD display, etc.) in conjunction with pages, forms and other information provided by MTS **16** or other systems or servers. As discussed above, the system is suitable for use with the Internet, which refers to a specific global internetwork of networks. However, it should be understood that other networks can be used instead of the Internet, such as an intranet, an extranet, a virtual private network (“VPN”), a non-TCP/IP-based network, any LAN or WAN or the like.

[0023] According to one embodiment, each user system **12** and all of its components are operator configurable using applications, such as a browser, including program code run using a central processing unit such as an Intel Pentium® processor or the like. Similarly, MTS **16** (and additional instances of MTS's, where more than one is present) and all of their components might be operator configurable using application(s) including program code run using a central processing unit such as an Intel Pentium processor or the like, or multiple processor units. Program code for operating and configuring MTS **16** to intercommunicate and to process web pages and other data and media content as described herein is preferably downloaded and stored on a hard disk, but the entire program code, or portions thereof, may also be stored in any other volatile or non-volatile memory medium or device as is well known, such as a ROM or RAM, or provided on any media capable of storing program code, such as a compact disk (“CD”) medium, digital versatile disk (“DVD”) medium, a floppy disk, and the like. Additionally, the entire program code, or portions thereof, may be transmitted and downloaded from a software source, e.g., over the Internet, or from another server, as is well known, or transmitted over any other conventional network connection as is well known (e.g., extranet, VPN, LAN, etc.) using any communication medium and protocols (e.g., TCP/IP, HTTP, HTTPS, WAP, Ethernet, etc.) as are well known. It will also be appreciated that program code for implementing aspects of the system can be implemented in any programming language that can be executed on a server or server system such as, for example, in C, C++, HTML, Java, JavaScript, WML, any other scripting language, such as VBScript and many other programming languages as are well known.

[0024] It should also be understood that each user system **12** may include differing elements, for example, one user system **12** might include a user's personal workstation running Microsoft's Internet Explorer browser while connected to MTS **16** by VPN, another user system **12** might include a thin-client netbook (e.g., Asus Eee PC®) running the Opera browser while connected to MTS **16** through an extranet, and another user system **12** might include a PDA running a WAP-enabled browser while connected to MTS **16** over third-party cellular networks.

[0025] According to one embodiment, each MTS **16** is configured to provide web pages, forms, data and media content to user systems **12** to support the access by user systems **12** as tenants of MTS **16**. As such, MTS **16** provides security mechanisms to keep each tenant's data separate unless the data is shared. If more than one MTS **16** is used, they may be located in close proximity to one another (e.g., in a server farm located in a single building or campus), or they may be distributed at

locations remote from one another (e.g., one or more servers located in city A and one or more servers located in city B). Each MTS **16** may include one or more logically and/or physically connected servers distributed locally or across one or more geographic locations. Additionally, the term “server” is meant to include a computer system, including processing hardware and process space(s), and an associated storage system and database application (e.g., relational database management system (“RDBMS”)), as is well known in the art. It should also be understood that “server system” and “server” are often used interchangeably herein. Similarly, the databases described herein can be implemented as single databases, a distributed database, a collection of distributed databases, a database with redundant online or offline backups or other redundancies, etc., and might include a distributed database or storage network and associated processing intelligence.

[0026] FIG. **2** illustrates elements of MTS **16** and various interconnections in an exemplary embodiment. In this example, the network interface is implemented as one or more HTTP application servers **100**. Also shown is system process space **102** including individual tenant process space(s) **104**, a system database **106**, tenant database(s) **108**, and a tenant management process space **110**. Tenant database **108** might be divided into individual tenant storage areas **112**, which can be either a physical arrangement or a logical arrangement. Within each tenant storage area **112**, a user storage **114** might similarly be allocated for each user.

[0027] It should also be understood that each application server **100** may be communicably coupled to database systems, e.g., system database **106** and tenant database(s) **108**, via a different network connection. For example, one application server **1001** might be coupled via the Internet **14**, another application server **100x-1** might be coupled via a direct network link, and another application server **100x** might be coupled by yet a different network connection. TCP/IP is the currently preferred protocol for communicating between application servers **100** and the database system, however, it will be apparent to one skilled in the art that other transport protocols may be used to optimize the system depending on the network interconnect used.

[0028] In aspects, each application server **100** is configured to handle requests for any user/organization. Because it is desirable to be able to add and remove application servers from the server pool at any time for any reason, there is preferably no server affinity for a user and/or organization to a specific application server **100**. In one embodiment, therefore, an interface system (not shown) implementing a load-balancing function (e.g., an F5 Big-IP load balancer) is communicably coupled between the application servers **100** and the user systems **30** to distribute requests to the application servers **100**. In one aspect, the load balancer uses a least connections algorithm to route user requests to the application servers **100**. Other examples of load balancing algorithms, such as round robin and observed response time, also can be used. For example, in certain aspects, three consecutive requests from the same user could hit three different servers, and three requests from different users could hit the same server. In this manner, MTS **16** is multi-tenant, wherein MTS **16** handles storage of different objects and data across disparate users and organizations.

[0029] As an example of storage, one tenant might be a company that employs a sales force where each user (e.g., a salesperson) uses MTS **16** to manage their sales process. Thus, a user might maintain contact data, leads data, customer follow-up data, performance data, goals and progress data, etc., all applicable to that user's personal sales process (e.g., in tenant database **108**). In one MTS arrangement, since all of this data and the applications to access, view, modify, report, transmit, calculate, etc., can be maintained and accessed by a user system having nothing more than network access, the user can manage his or her sales efforts and cycles from any of many different user systems. For example, if a salesperson is visiting a customer and the customer has Internet access in their lobby, the salesperson can obtain critical updates as to that customer while waiting for the customer to arrive in the lobby.

[0030] While each user's sales data might be separate from other users' sales data regardless of the

employers of each user, some data might be organization-wide data shared or accessible by a plurality of users or all of the sales force for a given organization that is a tenant. Thus, there might be some data structures managed by MTS **16** that are allocated at the tenant level while other data structures might be managed at the user level. Because an MTS might support multiple tenants including possible competitors, the MTS, in one implementation, has security protocols that keep data, applications, and application use separate. Also, because many tenants will opt for access to an MTS rather than maintain their own system, redundancy, up-time and backup are more critical functions and need to be implemented in the MTS.

[0031] In addition to user-specific data and tenant-specific data, MTS **16** might also maintain system-level data usable by multiple tenants or other data. Such system-level data might include industry reports, news, postings, and the like that are sharable among tenants.

[0032] In certain aspects, user systems **30** communicate with application servers **100** to request and update system-level and tenant-level data from MTS **16**; this may require one or more queries to system database **106** and/or tenant database **108**. MTS **16** (e.g., an application server **100** in MTS **16**) automatically generates one or more SQL statements (a SQL query) designed to access the desired information.

[0033] FIG. **3** illustrates an example of an object represented as a main table **200** that holds data items for multiple tenants. In the particular example shown in FIG. **3**, the main table **200** (.account) represents a standard Account entity that holds account information for multiple organizations. As shown, main table **200** includes an organization ID column **201** and an account ID column **202** that acts as the primary key for main table **200**. For example, for organization #1, column **201** shows that Org ID="ood 1," and two account IDs are listed ("ood 1" and "ood 2"). Main table **200** also includes a plurality of data columns **203** containing other information about each row. For example, for account ID "ood1," information is stored about the account name (i.e., "Enterprise"). Main table **200** may also include column **209** that stores the user ID of the user that owns or created the specific account that is stored in that row.

[0034] The organization ID column **201** is provided to distinguish among organizations using the MTS. As shown, N different organizations have data stored in main table **200**. In an exemplary embodiment, the organization IDs in column **201** are defined as Char (15), but may be defined as other data types. In one embodiment, the first 3 characters of the organization ID is set to a predefined prefix, such as "ood", although another subset of characters in the organization ID may be used to hold such a prefix if desired.

[0035] In the particular example of FIG. **3**, where the table represents a standard entity, data columns **203** are predefined data columns, or standard fields, that are provided to the various organizations that might use the table. In the Account entity example described above, such standard fields might include a name column, a site column, a number of employees column and others as would be useful for storing account-related information. Each of the data columns **203** is preferably defined to store a single data type per column.

Integration Environment

[0036] FIG. **4** illustrates a MTS environment for integrating and managing information pulled from a social network, such as twitter, Facebook, or Yammer. As illustrated in FIG. **4**, social network **30** is accessible by MTS **16** via a network **14**. The members **40** of social network **30** also access social network **30** via network **14**. Social network **30** is able to receive a message from a first member **40** that is then transmitted to one or more members **40** of social network **30**. Messages may be stored in a data store **34**. An interface **32** may be provided in order to provide a user interface for members **40**. Interface **32** may also provide an API through which MTS **16** is able to retrieve the messages in data store **34** and transmit both new messages and message replies through the social network **30**. Members **40** may be able to access social network **30** using any conventional Internet-connected device, including but not limited to a personal computer, a cellphone, a portable computing device, etc.

[0037] In an example embodiment, social network **30** is the twitter social network. A member **40** of twitter can broadcast a message (i.e., tweet) to all members of the twitter social network who are following tweets by the member **40**. A member **40** of twitter can also send a targeted message to a single, identified member of twitter, or to a group of members.

[0038] In an example embodiment, the MTS periodically polls the social network for new transmissions of messages and pulls information related to each newly transmitted message; this information is stored in the MTS in the form of a Conversation object. In an example embodiment wherein social network **30** is twitter, the MTS polls twitter's search API on a regular basis and stores all search results as twitter Conversation objects. Each twitter Conversation object is stored as a record in an entity of the MTS. An exemplary embodiment of a Conversation object for twitter is described in the table below.

Example Embodiment of a Conversation Object for Twitter

Field	Data Type	Description
Twitter Auto Number	The Row Id	where Conversation the Conversation Id object is stored in an entity of the MTS.
Twitter Id	Text(100)	The Id of the (External Id) message on twitter- pulled back via the API.
Content	Text[140]	The actual contents of the message (e.g., RT @EBF2010 European Banking Forum event looking at how to drive revenue; customer solutions, operational efficiency, and risk management). For twitter, this field may be limited to 140 characters.
Published Date	Date Time	The date and time the message was originally tweeted on twitter.
Author Name	Text(255)	The text name of the person that sent the message.
Twitter Profile URL	Text(255)	A URL to the author's twitter profile.
Type	Picklist	The source of the message, e.g., @reply, direct message, and Search.
Status	Picklist	The status or disposition of the conversation. As users process each conversation, they will disposition the conversation as to what happened. Values include: no action, ignore, case created, promoted to knowledge, solved by community, promoted to lead, etc.
Avatar Icon	Text field	Shows the user's with URL to image on twitter. the avatar The icon may be image resized to a fixed standard size so that it fits nicely in a List View or related list. Typically this field will be displayed next to the user's name.
Case Id Lookup		This is a foreign key to the Case that is filled in for any twitter conversation that is promoted to a Case.
Contact Id Lookup		This is a foreign key to the Contact object. If the message is linked to an existing contact with a twitter username, this field should be set to the contact the message was associated with.
Lead Id Lookup		If the message is "promoted" into a lead, this should specify the field LeadId that the message was converted into.
Solution Id Lookup		A foreign key for the solution for any message that is promoted to a solution.
Twitter Search Setup Id		A foreign key to the twitter search setup record that created this twitter Conversation Object.
Owner Lookup		The person(s) or User or queue that is Group assigned to look at this twitter conversation; this may be specified in the twitter search setup where you specify the owner for the results.
Campaign Id Lookup		Link to Campaign record
In Reply to	Text(255)	Hidden field used to keep track of twitter threads
Use Id	Text(255)	Hidden field used to keep track of twitter threads
Solution Hyperlink Icon		to show that Created formula field a solution is attached to this twitter conversation
Case Created Hyperlink Icon		to show that formula field a case is attached to this twitter conversation
Lead Created Hyperlink Icon		to show that formula field a lead is attached to this twitter conversation

Integrating Information From A Social Network Into A MTS

[0039] FIG. 5 illustrates an example process flow **500** for one embodiment of an integration with the twitter social network A message (i.e., tweet) is transmitted via the twitter social network, so when the MTS polls twitter's search API, the newly transmitted message is detected (**505**). In one embodiment, a user can define message filters to refine the search results.

[0040] If the message is a response to an earlier outbound message related to an existing case (**510**), then the user may create a Conversation object in the MTS and associate it with the existing case (**515**). The user can then create a case comment using the message text (**520**), and the case can be closed if it is appropriate to do so (**525**).

[0041] If the message is not a response to an earlier outbound message related to an existing case (510), then the user may create a new Conversation object (530) in the MTS. The MTS may then search through existing contacts and/or leads to see if the message was sent by an existing contact (535). If so, the MTS may associate the Conversation object with the existing contact (540). In one embodiment, if more than one existing contact and/or lead appears in the search results, then one of the following actions may be taken: (1) no association will be created, or (2) associations are created with all existing contacts and/or leads appearing in the search results, or (3) an association will be created with the most recent contact or lead. If the message was not sent by an existing contact (535), the MTS may create a new Contact or Lead and associate it with the Conversation object (545). In one embodiment, the Contact or Lead object includes a custom field to store the twitter username for the new contact.

[0042] If the message is a search tweet (i.e., a publicly broadcast status update), the MTS may retrieve the prior conversation, if any (550). If the message is an @username tweet (i.e., a publicly accessible targeted message) or direct message tweet (i.e., a private message), the user may be able to determine that the content of the message warrants creation of a new case and create a new case (555) and search the knowledge base (560) to see if there is an existing solution or any further information. If there is, then the user may be able to respond to the message to communicate the solution or further information (565)—the MTS may post the response to the twitter API (570).

[0043] In one embodiment, the user may be able to view a plurality of messages in a list view, select particular messages, and take actions upon the selected messages in the list (575). The user may be able to escalate a selected message to a case and create a new case (555). The user may be able to use the selected message to create a new knowledge base article (580). The user may be able to use the selected message to create a new lead (585). Or, the user may be able to choose to ignore the selected message and mark it for deletion (590). In one embodiment, the user may be able to change the status of the selected message (e.g., “solved by community”) or change other metadata related to the selected message.

[0044] In one embodiment, when a new Conversation object is created based on a transmitted message, metadata related to the transmitted message is generated and stored along with the Conversation object in the MTS. In one embodiment, when a new Conversation object is created, it can be categorized in accordance with any of a number of aspects, including but not limited to the message content, the author of the message, the recipient of the message, the date and/or time of the message, the case or lead or knowledge base article with which the message is associated, etc.

[0045] In one embodiment, messages from certain users (e.g., usernames belonging to MTS system administrators) may be filtered out in the search results.

User Interface

[0046] In one embodiment, the search for newly transmitted messages may be triggered manually. In one embodiment, the search for newly transmitted messages may be run automatically, possibly on a periodic basis. In one embodiment, the user can enable auto-creation of cases, leads, and/or contact

[0047] In one embodiment, outbound messages destined for the social network can be generated from several locations in the user interface—when the user is viewing a case, a Conversation object, a contact, and/or a lead. In one embodiment, the user interface allows creation of an outbound message from within the MTS user interface. In one embodiment, the user interface allows creation of an outbound message directly from the social network's user interface.

[0048] In one embodiment, once an action has been taken on a message (e.g., create case, create lead, ignore), the user interface indicates that an action has been taken on the message; in one embodiment, the user interface indicates the exact disposition of the message.

[0049] In one embodiment, reports can be generated to determine various statistics, which may include: how many messages were pulled in by the search, how many messages were reviewed, how many messages were promoted to cases or leads or knowledge base articles, how many

messages were ignored, and how many messages are left to be dealt with.

Setup and Administration

[0050] In one embodiment, a Search object is created in order to define various search terms for use when searching for messages in the social network. In one embodiment, multiple instances of Search objects may be created. Each search instance may have an associated owner (either username or queue). All conversations that are pulled in as a result of the Search object may be assigned to the owner associated with the search object. For example, conversations pulled in using the search terms “comcast issue”+“comcast problem” are assigned to the owner “twitter service.” In another example, conversations pulled in using the search terms “Verizon offer”+“which cable company” are assigned to the owner “twitter sales.”

Example Embodiment of a Search Object for Twitter Integration

TABLE-US-00002 Twitter Search AutoNumber The Row Id for the TermId twitter search in salesforce. Twitter Keyword Text Area(255) The Twitter search Search term Owner Lookup Lookup to User/Queue. All incoming twitter conversations from this search will be assigned to this queue. Unknown Picklist [Do If “Create Contact”, twitter nothing, Contacts will usernames Create Contact, automatically be Create created for unknown Lead] twitter users associated with Conversations. If “Create Lead”, Leads will automatically be created for unknown twitter users associated with Twitter Conversations. If Do nothing”, no object is created for unknown twitter users associated with Twitter Conversations. If the twitter user exists as a known contact then it should not create a contact or lead Lead Association Picklist Options: Setting Twitter Max Twitter ID Text(255) Hidden field Twitter Max Message Text(255) Hidden field ID Twitter Max Search Id Text(255) Hidden field Twitter Account ID Text(255) Hidden field Last Run Status Picklist Hidden field Error Msg TextArea(255) Hidden field Next API Reset Time Date/Time Hidden field Twitter Max In Text(255) Hidden field Progress Id

[0051] In one embodiment, an Account object is created in order to enable system administrators to keep track of necessary user account information, wherein the user account is related to the social network.

Example Embodiment of an Account Object for Twitter Integration

TABLE-US-00003 Twitter Account Id AutoNumber The Row Id for the twitter account in salesforce. Username Text (100) twitter username Password Text(50) twitter password Owner Lookup User/Queue lookup: this is the owner of the twitter account. It can be a user or a queue If private, then this can be used to control visibility to the account. Enable Auto-Case Checkbox When checked, Cases Creation and Contacts will automatically be created for incoming twitter replies and direct messages. Unknown twitter Picklist If “Create Contact”, usernames [Do nothing Contacts will automatically Create Contact, be created for unknown Create Lead] twitter users associated with, twitter conversations. If “Create Lead”, Leads will automatically be created for unknown twitter users associated with Twitter Conversations. If “Do nothing”, no object is created for unknown twitter users associated with Twitter Conversations. If the twitter user exists as a known contact then it should not create a contact or lead Bit.1y username Text(50) Username needed to create the shortened URL Bit.1y password Text(100) Stores the apiKEY to access the bit.1y API and generate the shortened URL.

[0052] While the invention has been described by way of example and in terms of the specific embodiments, it is to be understood that the invention is not limited to the disclosed embodiments. To the contrary, it is intended to cover various modifications and similar arrangements as would be apparent to those skilled in the art. Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

Claims

1-20. (canceled)

21. A system comprising: a server system comprising one or more processors, the server system configurable to cause: displaying, in a user interface (UI) on a display device, data of an account record stored in a customer relationship management (CRM) database system; identifying one or more social media posts displayable in the UI; communicating, responsive to the identifying, the one or more social media posts to the CRM database system; assigning, in the CRM database system, one or more social media users associated with the one or more social media posts, including mapping the one or more social media users to a CRM record in the CRM database system; providing an interface element in the UI for interacting with the one or more social media users; and tracking, using the CRM record in the CRM database system, interactions from the interface element.

22. The system of claim 21, wherein the interactions include a response to a post in an external social media network.

23. The system of claim 22, wherein the CRM record includes data from the response.

24. The system of claim 21, wherein the interface element is provided at least in part by the CRM database system.

25. The system of claim 21, wherein the account record is the CRM record.

26. The system of claim 21, wherein the mapping of the one or more social media users to the CRM record in the CRM database system includes matching the one or more social media users to a CRM contact stored in the CRM database system.

27. The system of claim 21, the server system further configurable to cause: triggering one or more follow-up actions, including creating a service request in the CRM database system, the one or more follow-up actions being trackable using the CRM database system.

28. A non-transitory computer-readable medium storing computer program instructions capable of being executed by one or more processors, the instructions configurable to cause: displaying, in a user interface (UI) on a display device, data of an account record stored in a customer relationship management (CRM) database system; identifying one or more social media posts displayable in the UI; communicating, responsive to the identifying, the one or more social media posts to the CRM database system; assigning, in the CRM database system, one or more social media users associated with the one or more social media posts, including mapping the one or more social media users to a CRM record in the CRM database system; providing an interface element in the UI for interacting with the one or more social media users; and tracking, using the CRM record in the CRM database system, interactions from the interface element.

29. The non-transitory computer-readable medium of claim 28, wherein the interactions include a response to a post in an external social media network.

30. The non-transitory computer-readable medium of claim 29, wherein the CRM record includes data from the response.

31. The non-transitory computer-readable medium of claim 28, wherein the interface element is provided at least in part by the CRM database system.

32. The non-transitory computer-readable medium of claim 28, wherein the account record is the CRM record.

33. The non-transitory computer-readable medium of claim 28, wherein the mapping of the one or more social media users to the CRM record in the CRM database system includes matching the one or more social media users to a CRM contact stored in the CRM database system.

34. The non-transitory computer-readable medium of claim 28, the instructions further configurable to cause: triggering one or more follow-up actions, including creating a service request in the CRM database system, the one or more follow-up actions being trackable using the CRM database system.

35. A computer-implemented method comprising: displaying, in a user interface (UI) on a display

device, data of an account record stored in a customer relationship management (CRM) database system; identifying one or more social media posts displayable in the UI; communicating, responsive to the identifying, the one or more social media posts to the CRM database system; assigning, in the CRM database system, one or more social media users associated with the one or more social media posts, including mapping the one or more social media users to a CRM record in the CRM database system; providing an interface element in the UI for interacting with the one or more social media users; and tracking, using the CRM record in the CRM database system, interactions from the interface element.

36. The computer-implemented method of claim 35, wherein the interactions include a response to a post in an external social media network.

37. The computer-implemented method of claim 36, wherein the CRM record includes data from the response.

38. The computer-implemented method of claim 35, wherein the interface element is provided at least in part by the CRM database system.

39. The computer-implemented method of claim 35, wherein the account record is the CRM record.

40. The computer-implemented method of claim 35, wherein the mapping of the one or more social media users to the CRM record in the CRM database system includes matching the one or more social media users to a CRM contact stored in the CRM database system.
